

Question

A、**B**、**C**、**D** are four non-metallic simple substances, where the corresponding elements of three of them are located in the same period, and the corresponding elements of two of them are located in the same group. **A** reacts with oxygen under high temperature conditions to obtain oxide **E**. **E** reacts with urea under high temperature conditions, or **E** reacts with CaB_6 under high temperature conditions and a nitrogen atmosphere to obtain a binary compound **F**, and **F** can be used as a component material of rocket engines. At -45°C , **C** and **D** react in CFCl_3 to obtain a yellow solid **G**. At -78°C , **D** and **G** react in CFCl_3 to obtain **H**. Heating **H** to above 0°C causes decomposition, yielding **D** and **I**. **I** can also be prepared by directly heating **C** and **D**. Mixing **C** and **I** at 90°C , and then heating sufficiently, yields **J**. **J** decomposes at temperatures above 450°C , yielding equimolar amounts of **C** and **I**. At low temperatures, **F** reacts with **H** to obtain **K** and a common Lewis acid **L**. **K** decomposes to form **B** and **D**.

Please provide the chemical formulas of each substance and choose the correct option.

- A.** All other options are incorrect.
- B.** Element **A** is a gas at room temperature.
- C.** In the process of generating **F**, the roles of urea and CaB_6 are completely consistent.
- D.** **E** is an amphoteric oxide
- E.** The central atom of **J** is sp^3d hybridized.
- F.** **F** and **H** react in a 1:1 ratio.
- G.** The boiling point of **I** is higher than **J**.

H. **A, B, C, D**, the elements corresponding to the four kinds of simple substances all have only one stable hydride.

Answer

Correct Answer: **G**

Detailed Explanation

Since **L** is a common Lewis acid and the elements involved are all non-metallic elements, it can be known that it should contain boron and halogen elements. Moreover, it is generated by the reaction of **H** and **F**, and **G**, **J**, and **I** have the same constituent elements as the compound of **H**. It does not conform to the properties of boron, so **F** should contain boron. Next, analyze the preparation process of the binary compound **F**: One way is the reaction of oxide **E** with urea. Since there is no other source of boron other than **E**, it indicates that it is boron oxide B_2O_3 , and the role of urea in the reaction is usually a carbon source or a nitrogen source, and it is more reasonable to act as a nitrogen source here; another way is the reaction of B_2O_3 with CaB_6 in a nitrogen atmosphere at high temperature. Combining the reaction conditions, it can be known that the existence form of the Ca element in the product is CaO , then the corresponding binary compound is a boron-nitrogen compound. In summary, starting from multiple conditions, it can be concluded that **F** should be BN . BN has a high melting point and high hardness, which is also consistent with the statement in the question "used in rocket engines".

In the process of generating BN , urea acts as a nitrogen source, and CaB_6 acts as a reducing agent (and also as a partial boron source), so the functions are different, and option C is incorrect.

B_2O_3 is a typical acidic oxide, and option D is incorrect.

CHECKPOINT

1 PTS

The chemical formula of compound **E** is B_2O_3

CHECKPOINT

1 PTS

The chemical formula of compound **F** is BN

CHECKPOINT

1 PTS

In the process of generating BN , the role of urea is a nitrogen source

CHECKPOINT

1 PTS

In the process of generating BN , the role of CaB_6 is a reducing agent and partial boron source

CHECKPOINT

1 PTS

B_2O_3 is a typical acidic oxide

The compound **E** is generated by the reaction of the element **A** with oxygen, indicating that **A** is elemental boron, and elemental boron is solid at room temperature, so option B is incorrect.

CHECKPOINT

1 PTS

A is elemental boron, which is solid at room temperature

Since **F** is BN , **L** generated by its reaction with **H** does not contain nitrogen, so **K** should contain nitrogen, and one of **B** and **D** generated by its decomposition should be nitrogen gas. **D** and **C** can generate four compounds with different ratios (**G**, **H**, **I**, **J**) under different conditions, which does not conform to the properties of nitrogen, so the element **B** is nitrogen gas.

CHECKPOINT

1 PTS

The chemical formula of **B** is N_2

Combined with the fact that three elements are in the same period, it is known that there are nitrogen and boron elements, so one of **C** and **D** must be a second-period element, and the element corresponding to **C** and boron form a common Lewis acid, which is a halogen. According to the question, **C** and **D** can form 4 binary compounds with different central element oxidation states. Only interhalogen compounds meet the conditions, which also conforms to the information "same group" mentioned in the question. Bromine can only generate two stable and confirmed binary compounds (BrF can only be observed in the spectrum, and the existence of BrF_7 is controversial), while iodine meets the conditions and can generate four fluorides, so the element corresponding to **D** is iodine, and correspondingly **C** is fluorine, and **L** is BF_3 .

CHECKPOINT

1 PTS

The chemical formula of **C** is F_2

CHECKPOINT

1 PTS

The chemical formula of **D** is I_2

CHECKPOINT

1 PTS

The chemical formula of **L** is BF_3

Finally, the four compounds of iodine and fluorine (**G**, **H**, **I**, **J**) should be IF , IF_3 , IF_5 , and IF_7 , respectively. According to the change in the ratio between the two elements that transform into each other (fluorine content **H** is less than **G**, **I** is greater than **H**, **J** is greater than **I** and only has two more fluorine atoms), it can be known that **G**, **H**, **I**, and **J** are IF_3 , IF , IF_5 , and IF_7 , respectively. The central atom of IF_7 is sp^3d^3 hybridized, and option E is incorrect.

CHECKPOINT

1 PTS

The chemical formula of compound **G** is IF_3

CHECKPOINT

1 PTS

The chemical formula of compound **H** is IF

CHECKPOINT

1 PTS

The chemical formula of compound **I** is IF_5

CHECKPOINT

1 PTS

The chemical formula of compound **J** is IF_7

CHECKPOINT

1 PTS

IF_7 has sp^3d^3 hybridization at the central atom

Finally, combining the fact that the decomposition of **K** will generate **B** and **D**, it can be known that the compound **K** is NI_3 . BN and IF react in a stoichiometric ratio of 1 : 3 to obtain BF_3 and NI_3 , so option F is incorrect.

CHECKPOINT

1 PTS

The chemical formula of compound **K** is NI_3

CHECKPOINT

1 PTS

BN and IF react in a stoichiometric ratio of 1 : 3 to obtain BF_3 and NI_3

For option G, IF_5 is more polar than IF_7 , and the orientation force is greater, and its boiling point is higher.

CHECKPOINT

1 PTS

IF_5 is more polar than IF_7 , and the orientation force is greater. Its boiling point is higher.

For option H, if hydrazoic acid is not considered, the element N has two stable hydrides, NH_3 and N_2H_4 , and the option is incorrect.

CHECKPOINT

1 PTS

If hydrazoic acid is not considered, the element N has two stable hydrides, NH_3 and N_2H_4

In summary, G is correct.